

Tajamul Ashraf

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RESEARCH INTERESTS

My research lies at the intersection of model generalization, multi-modal learning, learning from limited data (zero- and few-shot learning), and continual lifelong learning systems. I am particularly interested in building agentic systems that can actively perceive, reason, and interact with complex environments.

EDUCATION

Indian Institute of Technology Delhi, (IIT Delhi) New Delhi, India

Master of Science (M.S. by Research) in Information Technology June 2024

GPA: 9.28/10.0 Department Rank: 1st

Advisor: [Prof. Chetan Arora](#), Co-Advisor: [Prof. \(Dr.\) Krithika Rangarajan](#)

Thesis title: *Domain Adaptation in Breast Cancer Detection from Mammograms*.

National Institute of Technology Srinagar, (NIT Srinagar) Srinagar, India

Bachelors of Technology in Information Technology July 2022

GPA: 8.34/10.0

Advisor: [Prof. Janibul Bashir](#)

Thesis title: *Robust DL Models for Analyzing the Impact of Coherent Climatic Factors*.

RESEARCH EXPERIENCE

Nanyang Technological University (NTU) Singapore

Research Manager March 2026 - Present

Host Advisor: [Prof. Adams Wai Kin Kong](#)

- Working closely with the Interpol team on *video understanding and spatiotemporal modeling* for real-world surveillance scenarios.
- Contributing to AI-assisted systems that support **law enforcement investigations**, including the detection of harmful or suspicious interactions in sensitive cases (e.g., child safety and abuse content).
- Investigating multimodal and agentic-based techniques for robust temporal video reasoning.

King Abdullah University of Science and Technology (KAUST) Thuwal, Saudi Arabia

Research Intern Dec 2025 - March 2026

Host Advisor: [Prof. Bernard Ghanem](#)

- Introduced **MCTA**, a benchmark for evaluating multimodal clinical tool agents, focusing on grounded, multi-step reasoning over heterogeneous medical evidence.
- Developed evaluation protocols for *tool-augmented clinical reasoning*, assessing task decomposition, tool selection, and execution, and evidence-consistent decision making with an emphasis on reliability and interpretability.

Mohamed Bin Zayed University of Artificial Intelligence (MBZUAI) Abu Dhabi, UAE
Research Engineer Jan 2025 - Nov 2025
Host Advisor(s): [Prof. Rao Muhammad Anwer](#) and [Prof. Salman Khan](#)

- Developed **Agent-X**, a large-scale benchmark for evaluating deep multimodal reasoning in vision-centric agentic tasks, focusing on perception, grounding, and multi-step decision-making.
- Co-authored a comprehensive survey on **LLM post-training**, analyzing techniques for improving reasoning, alignment, and robustness in large language models.
- Proposed **MIRA**, a novel multimodal RAG framework for medical applications that fuses heterogeneous modalities to enable factual, adaptive, and clinically grounded agent behavior.
- Contributed to **CarePilot**, a multi-agent system for long-horizon healthcare task automation, enabling coordinated planning and execution in complex clinical workflows.

Microsoft Research (MSR) Bangalore, INDIA
Research Intern July 2024 - Dec 2024
Host Advisor(s): [Dr. Mohit Jain](#) and [Dr. Prachi Jain](#).

- Co-introduced **Cataract-MSICS** dataset and proposed a phase-conditioned method with pseudo-labeling for surgical tool segmentation.
- Worked on Knowledge Graph-based Information Retrieval with RAG.
- Automated LLM pipelines with a semi-structured knowledge base (SKB) for medical chatbots.

AI & Robotics Technology Park, I-Hub (ARTPARK) Bangalore, INDIA
Visiting Student Dec 2021 - May 2022
Advisor: [Prof. Raghu Krishnapuram](#).

- Developed an automated washroom cleaning robot **RoboJanitor**.
- Designed robust models for pose estimation, detection, classification, and SLAM.
- Developed 3D pose models and REST APIs for production.

Indian Institute of Science (IISc) Bangalore, INDIA
Research Intern Mar 2021 - Sep 2021
Advisor(s): [Prof. S.N. Omkar](#) and [Prof. Balaji Prabu](#).

- Developed a novel human pose comparison model to assist yoga teachers during online classes.
- Utilized BlazePose to accurately mark key points of both teachers and students during live sessions.
- Designed a feedback mechanism to compare poses and provide real-time corrections to students.

WORK EXPERIENCE

LEENA AI Remote(WFH), USA
Software Developer I May 2022 - Oct 2022
Line Manager: [Anand Prajapati](#)

- Executed full-stack software development for the Product and ML team.
- Worked on cross-functional projects, architecting solutions to meet client requirements.

BSNL ALTTC Delhi, INDIA
Cybersecurity Trainee Jan 2019 - Mar 2019
Line Manager: [R.K. Pandit](#)

- Troubleshoot and resolved network issues related to Wi-Fi and Bluetooth.

- Implemented security measures and optimized network performance in wireless environments.

TEACHING and ACADEMIC SERVICES

Teaching Assistant | Nanyang Technological University, Singapore

- Probability and Statistics (SC2000) | Spring 2026 with [Prof. Adams Wai Kin Kong](#)

Teaching Assistant | MBZUAI, Abu Dhabi

- Trustworthy Medical Vision (CV8502) | Fall 2025 with [Prof. Yutong Xie](#)

Teaching Assistant | Indian Institute of Technology, Delhi

- Computer Vision (COL780) | Spring 2024 with [Prof. Chetan Arora](#)
- Computer Vision for Robotics (JRL780) | Spring 2024 with [Prof. Chetan Arora](#)
- Computer Science Fundamentals and Programming (COL100) | Fall 2023 with [Prof. Chetan Arora](#) and [Prof. Vireshwar Kumar](#)
- Special Topics in Machine Learning (COL870) | Spring 2023 with [Prof. Anurag Mittal](#)
- Data Structures and Algorithms (COL106) | Fall 2022 with [Prof. Naveen Garg](#) and [Prof. Ashish Chiplunkar](#)

Conference/Journal Activities | Reviewer

- NeurIPS'26 | ICCV'25 | ACM MM'26, 25 | MICCAI'26, 25, 24 | WACV'25, 24 | CVIP'23, 22 | HPEC'23
- International Journal of Computer Vision (IJCV)
- IEEE Transactions on Artificial Intelligence
- Pattern Recognition
- IEEE Transactions on Parallel and Distributed Systems

Conference Volunteer | Organizer

- Program Committee Member for ACM Multimedia 2026 held in Rio de Janeiro, Brazil
- Volunteer for ICLR 2026 held in Rio de Janeiro, Brazil
- Student Volunteer for ACM COMPASS 2025 held in Toronto, Canada
- Served as a program volunteer for CVIP 2023 conference in IIT Jammu, INDIA

Invited Talk | Speaker

- Delivered a keynote talk on “*Multimodal Systems for Vision-Language domains in Advancing Healthcare*” at the BRAIN-AID Seminar, hosted by the College of Medicine, University of Lagos, Nigeria (April 2025).
- Presented two talks on Research Directions, Placements, and Internships to undergraduate IT students at the National Institute of Technology, Srinagar (June 2024).

Pilot Mentorship Program | Mentor

I actively mentor students across various levels, guiding them in academic projects, final year theses, and independent research initiatives, with a focus on helping them achieve strong technical outcomes and potential research publications.

Students (* denotes current)

8. [Burhaan Rasheed Zargar](#)* | NIT Srinagar | 2025-26
Thesis: *Language Translator and Text to Speech tool for Kashmiri*

7. [Ifrah Mushtaq](#), [Irtifa Aalam](#)* | Kashmir University, NIT Srinagar | 2025-26
Thesis: *Automatic Speech Recognition and Phonetic analysis for Kashmiri Language*
 6. [Mohammad Hashid](#)* | NIT Srinagar | 2025-26
Thesis: *Efficient Breast Cancer Classification using RL*
 5. [Abrar Riyaz](#), [Wasif Tak](#)* | NIT Srinagar, Thapar University | 2025-26
Thesis: *Robust tool segmentation in surgical scenarios*
 4. [Tawaheed Tarid](#), [Sonia Yadav](#)* | NIT Srinagar | 2025-26
Thesis: *Grounding-based Multi-object Tracking in Resource-Constrained Devices.*
 3. [Suhaib Salmani](#) | NIT Srinagar | 2024-25
Thesis: *VLM-based Masked Learning for Efficient Medical Object Detection.*
 2. [Rajes Manna](#) | NIT Srinagar | 2024-25
Thesis: *Source-Free Domain Adaptation using Adversarial Class-Aware Teacher.*
 1. [Asrar ul Haq](#) | NIT Srinagar | 2023-24
Thesis: *Transformer-Based Unsupervised Domain Adaptation in a Federated Setup.*
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HONORS and AWARDS

- ICLR 2026 Registration and Travel Grant (Rio de Janeiro, Brazil)
 - MICCAI 2025 Outstanding Reviewer Award
 - Awarded ICCV 2025 Travel Grant (Honolulu, Hawaii).
 - Awarded ACM Multimedia 2025 Travel Grant (Dublin, Ireland).
 - Awarded MBZUAI Travel Grant to attend Agentic-AI Summit 2025 at UCB, California.
 - Finalist, Agent-X LLM Agents MOOC Competition organized by UC Berkeley RDI.
 - Awarded ACM COMPASS 2025 Registration Grant held in Toronto, Canada
 - Secured 2nd place (AED 500 award) in the MBZUAI Department Logo Design Competition.
 - Awarded \$500 compute grant from alphaXiv for my contributions in LLM research.
 - Nominated as Diversity Intern by Microsoft Canada.
 - Honorable mention by IIT Delhi for research contributions during my Master's degree.
 - Awarded MICCAI 2024 Travel Grant.
 - Awarded IIT Delhi Endowment Grant 2024.
 - Won India's largest National Entrepreneurship Challenge at IIT Bombay.
 - Gold Microsoft Student Learn Ambassador for NIT Srinagar.
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TECHNICAL SKILLS

Languages: Python, C, C++, MATLAB, SQL, HTML/CSS, Javascript

Frameworks: PyTorch, TensorFlow, Keras, Scikit-Learn, OpenCV, HuggingFace, Flask

Developer Tools: Git, Docker, VS Code, Visual Studio, PyCharm, Linux

EXTRACURRICULAR and SOCIAL ACTIVITIES

Gold [Microsoft Student Learn Ambassador](#)

Actively promoted technology learning and fostered a community of student developers in Kashmir. Organized workshops, webinars, and hackathons to encourage student innovation and collaboration.

Founder [Ralith Milth](#)

Established an NGO focused on raising awareness and support for drug abuse prevention and rehabilitation. Spearheaded community outreach programs with local organizations to support affected individuals.

Co-founder [AT Talks](#)

Hosting a podcast discussing the intersection of technology and society, featuring expert insights. Managed guest interviews, content strategy, and promotion to grow the podcast's reach and impact.

MEDIA APPREANCES

Media & Outreach:

- **2025:** *A new stress test for AI agents that plan, look, and click.* [Read more](#)
- **2022:** *Meet a Microsoft Learn Student Ambassador graduate: Tajamul Ashraf.* [Read more](#)
- **2022:** *NIT Srinagar students win India's largest National Entrepreneurship Challenge.* [Read more](#)
- **2022:** *Hult Prize: A futuristic approach to entrepreneurship.* [Read more](#)
- **2021:** *Workshop on entrepreneurship, research and innovation at NIT Srinagar.* [Read more](#)

REPRESENTATIVE PUBLICATIONS

(* indicates equal contribution, † indicates my role as mentor)

C8 *CarePilot: A Multi-Agent Framework for Long-Horizon Computer Task Automation in Healthcare.*

 [Code](#)  [Paper](#)

A. Ghosh, **Tajamul Ashraf**, R. K. Singh, N. Saeed, S. Saha, X. Chen, S. Khan

CVPR 2026 (Conference on Computer Vision and Pattern Recognition), Findings

C7 *Agent-X: Evaluating Deep Multimodal Reasoning in Vision-Centric Agentic Tasks.*

 [Code](#)  [Paper](#)  [Data](#)

Tajamul Ashraf*, Amal Saqib*, Hanan Ghani, Muhra AlMahri, Yuhao Li, Noor Ahsan, Umair Nawaz, Jean Lahoud, Hisham Cholakkal, Mubarak Shah, Philip Torr, Fahad Shahbaz Khan, Rao Muhammad Anwer, Salman Khan

ICLR 2026 (International Conference on Learning Representations)

C6 *MIRA: A Novel Framework for Fusing Modalities in Medical RAG.*  [Code](#)  [Paper](#)

Jinhong Wang*, **Tajamul Ashraf*†**, Zongyan Han, Jorma Laaksonen, Rao Muhammad Anwer

MM 2025 (ACM Multimedia)

C5 *TITAN: Query-Token based Domain Adaptive Adversarial Learning.*  [Code](#)  [Paper](#)

Tajamul Ashraf, Janibul Bashir

ICCV 2025 (IEEE/CVF International Conference on Computer Vision)

- C4** *Phase-Informed Tool Segmentation for Manual Small-Incision Cataract Surgery.*  [Paper](#).
Bhuvan Sachdeva, Naren Akash, **Tajamul Ashraf**, Simon Mueller, Thomas Schultz, Maximilian WM Wintergerst, Niharika Singri Prasad, Kaushik Murali, Mohit Jain
MICCAI 2025 (Medical Image Computing and Computer Assisted Intervention)
- C3** *D-MASTER: Mask Annealed Transformer for Unsupervised Domain Adaptation in Breast Cancer Detection from Mammograms.*  [Code](#)  [Paper](#).
Tajamul Ashraf, Krithika Rangarajan, Mohit Gambhir, Richa Gauba, Chetan Arora
MICCAI 2024 (Medical Image Computing and Computer Assisted Intervention)
- C2** *HF-Fed: Hierarchical based customized Federated Learning Framework for X-Ray Imaging.*
 [Code](#)  [Paper](#).
Tajamul Ashraf, Tisha Madame
MICCAI Workshop 2024 (Deep-Brea3th). Best Student Paper Award!
- C1** *TransFed: A way to epitomize Focal Modulation using Transformer-based Federated Learning.*
 [Code](#)  [Paper](#).
Tajamul Ashraf, Fuzayil Mir, Iqra Altaf Gillani
WACV 2024 (IEEE/CVF Winter Conference on Applications of Computer Vision)
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REFERENCES

Prof. Mubarak Shah 
Professor
UCF, USA

Mohit Jain 
Principal Researcher
Microsoft Research, India

Prof. Chetan Arora 
Professor
IIT Delhi, India